Design 3 Coursework

Building A

Group 11-
Ali Hassan
Ismail Mohammad
Haoyu Gong
Mahmoud Elgebaly

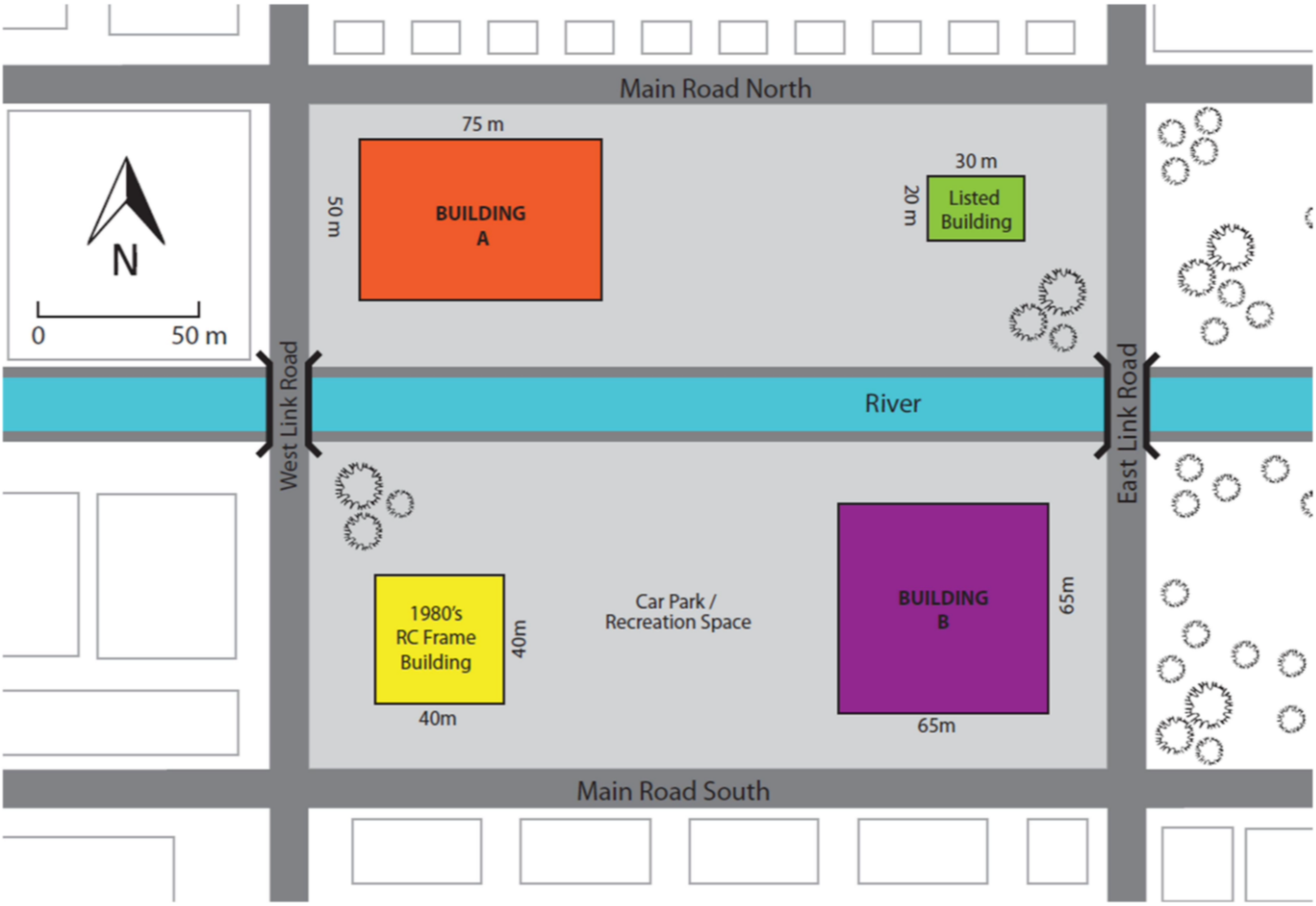

Figure 1: Layout of the Proposed Development

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1: Introduction

1.1: Project Brief

The project's objective is to develop a structural design that satisfies the client's needs while taking the site's unique constraints into account. Several crucial considerations must be made to ensure the design merges with the environment and cause the least amount of disturbance. Important elements affecting the design are:

Location: Originally a railway yard and maintenance depot, the property is on the outskirts of the city centre. Due to traffic limits during rush hour and the major main roads that border it, construction operations are limited. To prevent delays and interruptions, deliveries and other tasks will need to be carefully planned to take place outside of busy hours.

Adjacent Infrastructure: A river runs through the site's reinforced concrete walls, and several buildings, including an office block from the 1980s and a listed Victorian locomotive depot, must continue to function throughout development. The design and construction stages must be carefully handled to prevent damage to these buildings from interfering with their use.

Historical Background: Because of the site's lengthy history as a railway yard, the stability of the new buildings may be impacted by subterranean structures or contaminants. A comprehensive geotechnical study is required to ensure the earth can sustain the new structures and to find any hazards.

The design and construction team will create a solution that minimises interruption, fits well within its environment, and guarantees the project is finished swiftly by taking these elements into account.

2: Floor Options

2.1: Approach Taken

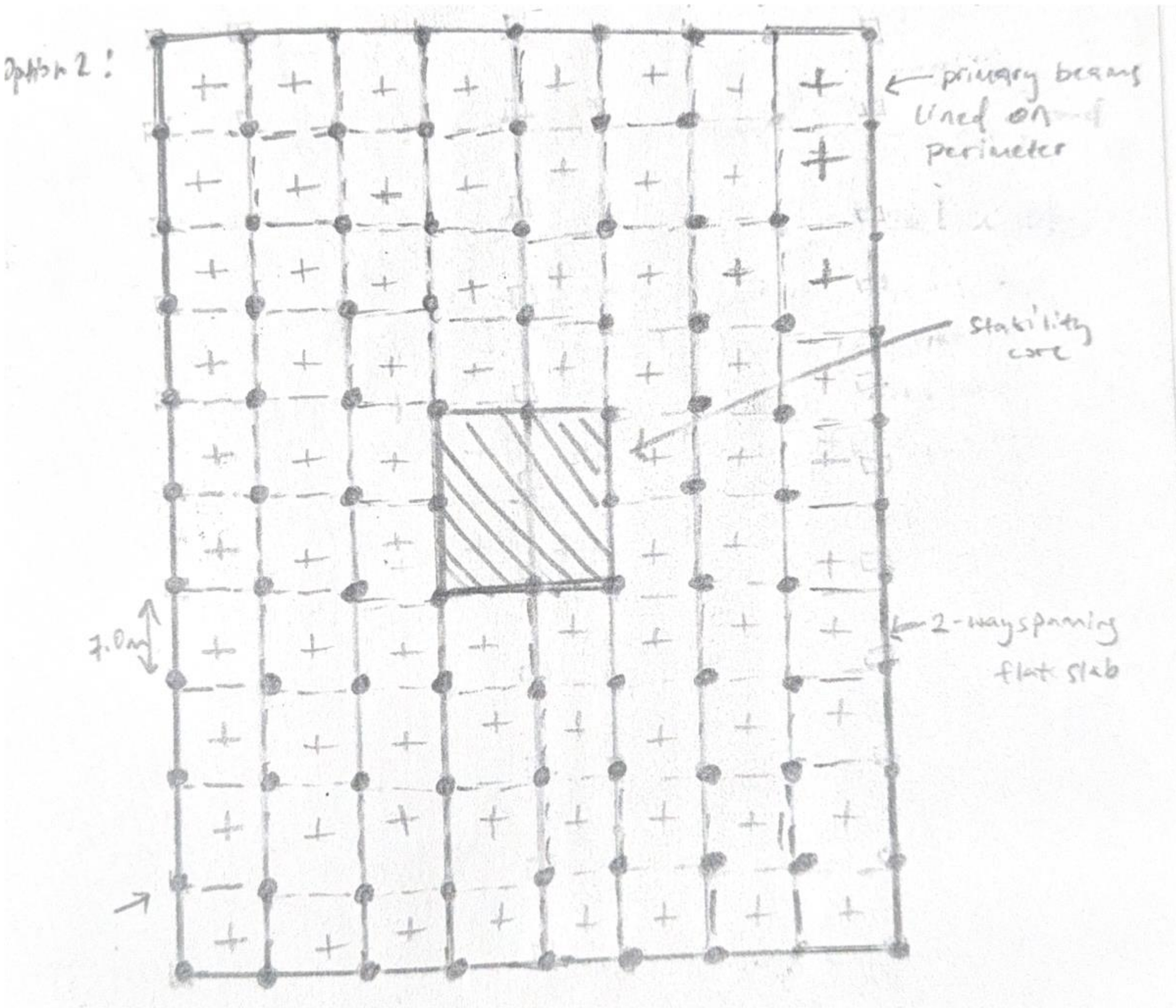
For Building A, selecting the optimal structural frame solution involves a thorough evaluation of two options: steel and concrete. A comprehensive appraisal will be conducted, utilising both qualitative and quantitative methods to assess each option against the project's key considerations. The qualitative analysis will explore the strengths and weaknesses of each framework, while the quantitative evaluation will apply a scoring system to compare the options objectively.

The assessment will be based on the following important factors, identified from the client's requirements and standard civil engineering practices:

- Carbon footprint
- Visual appeal
- Ease of Construction
- Cost
- Fire resistance
- Safety
- Flexibility
- Thermal performance
- Maintenance requirements

A final table will be produced, summarising how each frame performs across all factors. The table will score the frame solutions from 0-10 based on their performance on each factor. This ranking will guide the selection of the most appropriate structural solution for Building A, ensuring the decision is based on a clear, structured analysis of all relevant factors.
This approach ensures transparency and objectivity, providing all stakeholders with a comprehensive overview of the decision-making process. By presenting the results in a clear and organised manner, it will be easier to identify the most suitable option and move forward with confidence in the selected solution.

2.2: Concrete Frame Solution



Slab Design

The concrete frame option relies on two-way spanning slabs, physically solid; however, the cement production process is one of the biggest sources of CO_2 emissions in the building sector, resulting in a greater carbon footprint. This system distributes the loads in both directions, transferring them directly to the supporting columns without the need for secondary beams. The slab thickness of 250 mm is adequate to handle the imposed loads of 4.0 kN/m² for the upper floors and 5.0 kN/m² for the ground floor. Concrete is naturally fire-resistant, so this solution easily meets the 1-hour fire resistance requirement set out by the client.

- Advantages: The two-way slab system simplifies construction by eliminating the need for secondary beams, while its natural fire resistance reduces additional fireproofing costs. Concrete also offers long-term durability, making it an attractive option for longevity.
- Disadvantages: Concrete slabs are heavier than steel decking solutions, leading to increased foundation loads and more costly foundations. Additionally, concrete construction requires significant on-site work and long curing times, which can slow down the project's timeline. (*Steel vs. Concrete: Comparing the Pros and Cons in Construction | Solid State Steel*, 2023)

Beam Design

Although the two-way slab system minimises the need for internal beams, perimeter beams are still required to support the edges of the slab. These beams will span 7.5 meters, with a depth of 350-375 mm, ensuring that loads from the slab are transferred efficiently to the columns.

- Advantages: Perimeter beams help distribute loads to the supporting columns without disrupting the open-plan space inside the building.
- Disadvantages: These beams add to the structural complexity and increase the overall weight, putting more pressure on the foundation. This could affect the project's cost and environmental footprint. (Natsoulis, 2018) (*Steel vs. Concrete: Comparing the Pros and Cons in Construction | Solid State Steel*, 2023)

Column Design

The columns in the concrete frame solution are reinforced concrete columns spaced 7.5 meters apart. These columns must support the significant loads from the slabs and beams, particularly on the lower floors. Columns on the ground floor will be 450 mm x 450 mm, reducing to 400 mm x 400 mm on the upper floors.

- Advantages: Concrete columns provide excellent compressive strength and inherent fire resistance, reducing additional safety costs.
- Disadvantages: Concrete columns are bulkier than steel columns, reducing usable space and potentially affecting the building's aesthetic design. (*Steel vs. Concrete: Comparing the Pros and Cons in Construction | Solid State Steel*, 2023)

Lateral Stability

The concrete frame achieves lateral stability through a central stability core, which acts as a shear wall. This core is designed to resist lateral forces such as wind loads, transferring them safely to the foundation.

- Advantages: Central cores provide strong lateral resistance and double as vertical service zones, efficiently combining structural and functional purposes.
- Disadvantages: The centralised core increases the overall weight of the building, requiring more robust foundations and adding complexity to the construction process. (Natsoulis, 2018) (*Steel vs. Concrete: Comparing the Pros and Cons in Construction | Solid State Steel*, 2023)

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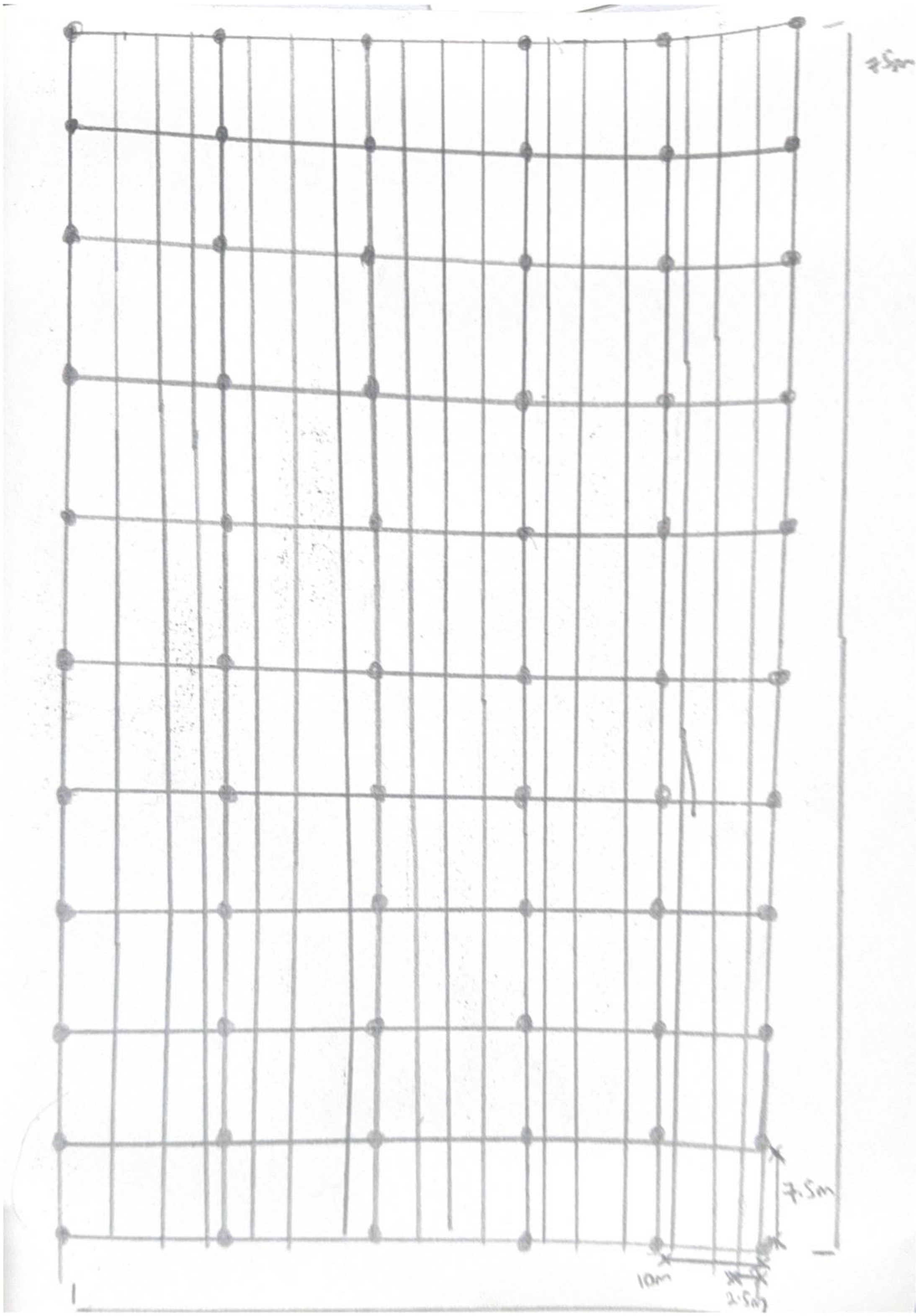
2.3: Steel Frame Solution



Slab Design

The steel frame solution uses a composite metal decking system, which consists of steel decking with concrete poured on top. The composite action between the concrete and steel decking allows for thinner slabs and smaller beams compared to a concrete frame. Furthermore, its lightweight also minimises material usage, meaning there's a reduction in the overall carbon footprint of the project. The reduced material contributes to less embodied carbon compared to traditional concrete slabs, which are heavier and require more material. Finally, the slab thickness for this system will range from 130-150 mm, ensuring that it can handle the required loads while minimising the structural depth.

- Advantages: The lightweight composite system reduces the overall load on the building's foundation and speeds up construction, as the decking is prefabricated and can be rapidly installed. (Valle, 2021) (*Steel vs. Concrete: Comparing the Pros and Cons in Construction | Solid State Steel*, 2023) The slab also provides ample space for integrating services like HVAC and electrical systems, reducing coordination challenges on-site.
- Disadvantages: The steel components of the decking system require additional fireproofing, such as intumescent coatings, to meet fire resistance standards.

Beam Design

The steel frame will use universal steel beams (both primary and secondary beams) along with the composite decking system. Strong and adaptable, universal beams provide steady strength in both vertical and horizontal directions. Because of this, they can move large objects while keeping the area open. The main beams, which are intended to sustain the necessary loads, will extend 7.5 meters and have a depth of 450–600 mm. 5m long secondary universal beams will be used to control deflection across the slabs.

- Advantages: Universal beams are reasonably priced and widely accessible. Because of their regular sizes, they are easy to work with and have a great load-bearing capability, which speeds up building. Additionally, they are compatible with composite decking systems.
- Disadvantages: Fireproofing steel beams, including universal ones, raises the initial cost. Nonetheless, the quicker installation and easier access to materials contribute to a reduction in the entire project's time and expenses.

Column Design

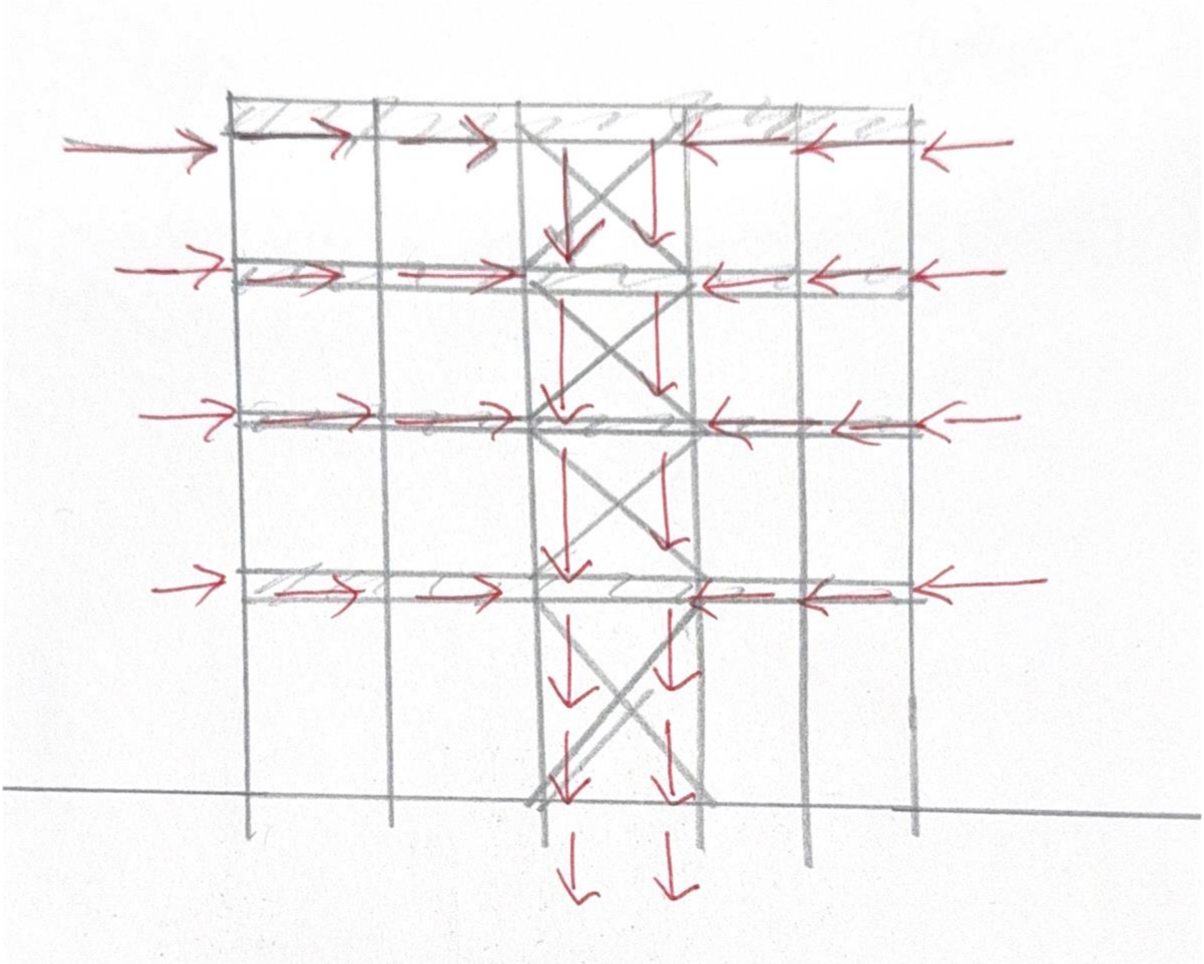
The steel frame system's 7.5-meter-distance steel columns will hold up the main beams. The size of these steel columns will reduce as the weight lowers higher up the building; they will be 450 mm x 450 mm on the lower levels and 400 mm x 400 mm on the upper floors.

- Advantages: Steel columns provide more usable space inside the building since they are thinner than reinforced concrete columns. Because they are prefabricated, they may also be installed more quickly. While steel offers exceptional strength and flexibility, contemporary fireproofing techniques guarantee that the columns adhere to safety regulations.

Lateral Stability

Lateral stability in the steel frame is achieved through vertical cross-bracing along the middle and corners of all sides of Building A. The cross-bracing will be located on the ground and second floors, designed to resist wind loads and seismic forces. Horizontal bracing is unnecessary, as vertical cross-bracing is sufficient to manage lateral forces. (*Steel vs. Concrete: Comparing the Pros and Cons in Construction | Solid State Steel*, 2023)

- Advantages: Steel cross-bracing is lightweight, easy to install, and provides excellent lateral stability without adding excessive weight to the structure.



2.4: Score Table

Factors	Steel Frame	Concrete Frame
Carbon footprint	Steel is recyclable, and this reduces waste during demolition. However, its production is energy-intensive, primarily due to the processes involved in smelting.. (8/10)	Concrete has a significantly larger carbon footprint, mainly because of the energy-intensive cement production process. Concrete is less sustainable at the end of its life as it is harder to recycle. (5/10)
Visual appeal	Steel offers modern, sleek aesthetics, ideal for open-plan designs. The ability to create larger spans with fewer columns gives architects more freedom, allowing for cleaner, more minimalist structures. (10/10)	Concrete provides a more solid, heavy appearance, which can suit some architectural styles but may limit flexibility in creating open spaces. (5/10)
Ease of Construction	The prefabricated nature of steel allows for rapid on-site installation. Since steel is less affected by weather conditions, construction timelines are more predictable, and projects can be completed faster. (9/10)	Concrete requires on-site work such as formwork and curing. Curing times, particularly, extend the construction period, and weather conditions can slow down the process further making it not easy. (3/10)

Cost	Steel has a higher upfront cost, especially with the need for fireproofing. However, the quicker construction and reduced labour costs can offset the higher material costs, making steel more cost-effective in the long run. (7/10)	Concrete is cheaper upfront in terms of material costs, but the extended construction time due to curing and labour-intensive processes can lead to higher overall labor costs. (6/10)
Fire resistance	Steel needs fireproofing, as it loses strength when exposed to high temperatures. Without these measures, steel can pose safety risks in case of a fire. (6/10)	Concrete is naturally fire-resistant and doesn't require additional treatment to maintain structural integrity during a fire. This makes concrete require no additional fireproofing. (9/10)
Safety	Steel is strong and safe under normal conditions, particularly with the addition of fireproofing and corrosion protection. It is also flexible under dynamic loads such as wind and earthquakes, making it a reliable option for various building types. However, maintenance is crucial to ensuring long-term safety. (8/10)	Concrete is inherently safe with its high compressive strength, making it ideal for large and heavy buildings. It also naturally resists fire and most environmental degradation, reducing the long-term risk of structural failures. However, it is less flexible under dynamic loads. (7/10)
Flexibility	Steel offers high flexibility in terms of design. It supports larger spans with fewer columns, making it ideal for open-plan spaces. Asymmetric beams can optimize space usage by reducing beam size while maintaining structural integrity. (9/10)	Concrete requires thicker columns and slabs to support the same loads, which reduces its flexibility in design. This can limit the creation of large open spaces and make future modifications more difficult. (5/10)
Thermal Performance	Steel does not naturally retain or release heat well, which can lead to higher heating and cooling costs. Additional insulation or cladding is required to improve energy efficiency, adding to construction complexity and cost. (6/10)	Concrete offers excellent thermal mass, helping to regulate indoor temperatures by absorbing heat during the day and releasing it at night. This contributes to energy efficiency. (8/10)
Maintenance Requirements	Steel needs regular maintenance, particularly in environments prone to moisture or corrosion. Protective coatings can reduce the frequency of maintenance, but inspections are required to ensure the steel's longevity. Fireproofing materials may also need periodic checks and replacement. (8/10)	Concrete is low maintenance, with fewer concerns regarding corrosion or environmental degradation. However, cracks should be monitored and repaired to prevent long-term structural issues. Overall, concrete requires less frequent maintenance than steel structures. (9/10)
Total	71/90	57/90

Further qualitative appraisal on Carbon Footprint:

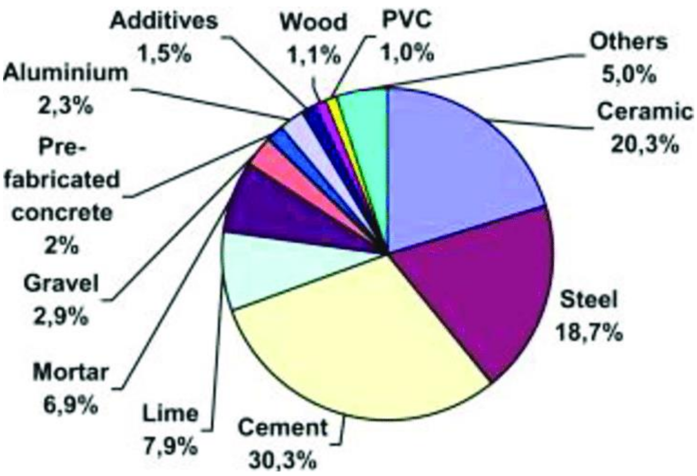


Figure 2: CO2 contribution from various construction materials (Hermawan et al., 2015)

According to the pie chart, steel produces 18.7% of CO2 emissions, while cement contributes the most, at about 30.3%. This demonstrates how cement, an essential part of concrete, greatly raises the carbon footprint of the building. Steel, on the other hand, has significant emissions during production, but throughout the building's life, its recyclable nature makes it more environmentally beneficial. Therefore, despite its initial environmental cost during manufacture, steel might provide a superior option when looking at the increase in carbon footprint using concrete.

2.5: Conclusion: Steel Frame solution is chosen

The steel frame system was finally selected for Building A because of its substantial benefits in terms of flexibility, material efficiency, and construction speed. In contrast to the labour-intensive and time-consuming concrete curing process, prefabricated steel components may be quickly installed on-site, minimising disruption and guaranteeing a speedier project timeline. By enabling smaller beam sizes and thinner slabs, the combination of asymmetric beams and composite metal decking further improves material efficiency while lowering the overall weight of the building and the need for a foundation. Additionally, this design accommodates expansive open-plan areas with fewer internal columns, meeting the client's demand for flexible office and studio layouts. Although steel needs extra fireproofing, it's more economical in the long run, recyclable, and environmentally friendly to use recycled steel instead of concrete, which has a larger carbon footprint because of the production of cement. Therefore, from a qualitative standpoint, steel is the more sustainable choice due to its capacity to be recycled and its ability to reduce waste at the end of its life. Finally, the choice of picking a steel frame as the final solution was proven by the scoring system created, which showed steel scoring 71/90 as compared to concrete with 57/90. This showcases the steel frame’s ability in front of the nine important factors considered.

3: Final Calculations

<u>Office Building (Building A):</u>

Roof:	Imposed	7.5 kN/m^2		
1st to 3rd floors:	Imposed	4.0 kN/m^2	Services	0.25 kN/m^2
Ground floor:	Imposed	5.0 kN/m^2	Services	0.25 kN/m^2

<u>Residential Building (Building B):</u>

Roof:	Imposed	7.5 kN/m^2		
1st to 5th floors:	Imposed	3.0 kN/m^2	Services	0.25 kN/m^2
Ground floor:	Imposed	5.0 kN/m^2	Services	0.25 kN/m^2

Floor selection:

The critical span is 7.5m, as the span between two columns. The characteristic load is 7.5kN/m^2, which can be estimated as 10kN/m^2 on the Bison Manufacturing Table (for Hollow Composite Floors). The load/span table gives the dimensions of the composite metal decking flooring and accounts for C30 concrete topping, which is within our design criteria. Reading from the load/span table, the span of the composite metal decking is found to be 7.8m, which is greater than 7.5m, and therefore passes the check.

Bison Manufacturing / 01283 817500

Hollow Composite Floors
Load/Span Table

A Bison Hollow Composite Floor is a Bison Hollowcore Slab with a structural concrete topping. This combination delivers increased structural performance overall, with an improved lateral load distribution where necessary for heavy point loads. The floor may be designed in the unpropped or partially propped condition to suit particular requirements. It is particularly suitable for industrial buildings, high buildings, car parks or other structures where additional longitudinal and transverse tying is required.

Overall structural depth mm	Unit Depth	Self wt kN / m²	Spans indicated below allow for characteristic service load (live load) plus self weight plus 1.5 kN/m² for finishes								
			Characteristic service loads kN/m²								
			0.75	1.5	2.0	2.5	3.0	4.0	5.0	10.0	15.0
			Effective span in metres								
200	150	3.6	8.25	8.25	8.1	7.9	7.7	7.4	7.1	5.8	4.9
250	200	4.2	10.4	9.9	9.7	9.4	9.2	8.8	8.4	7	6
300	250	4.5	11.7	11.2	10.9	10.6	10.4	9.9	9.5	7.8	6.8
375	300	5.8	14.5	14	13.7	13.5	13.2	12.7	12.3	10.7	8.8
425	350	6.2	16	15.5	15.2	14.9	14.6	14.1	13.7	11.9	10
475	400	6.6	17.1	16.5	16.2	15.9	15.6	15.1	14.6	12.7	10.7
525	450	7.2	18.5	17.9	17.6	17.3	17	16.4	15.9	13.9	11.7

The above data is based upon 50 or 75mm structural topping of C30 concrete which should be regarded as a minimum. Other topping depths may be recommended in some circumstances. Design data for alternative combinations are available from Bison design offices. In relation to the overall structural concept of the building it is also important also to consider topping reinforcement, daywork and movement joints.

Figure 3: Load Span table for floors

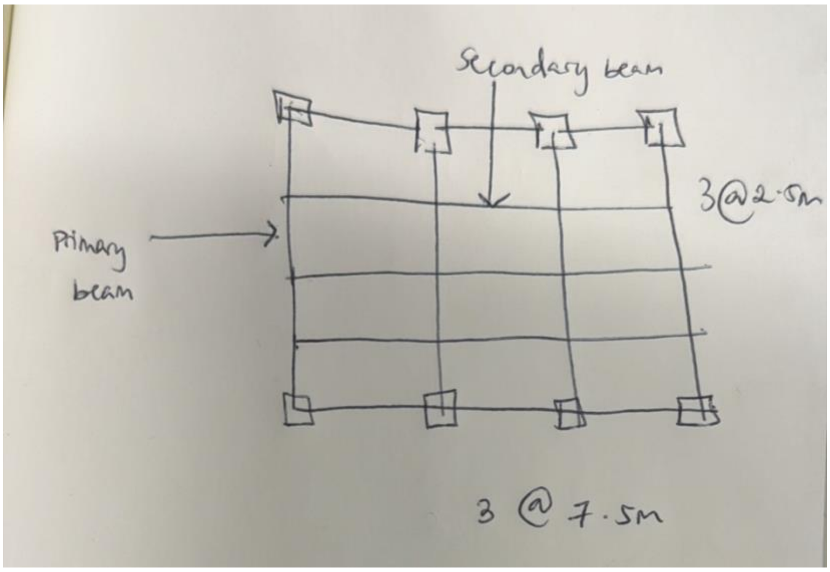
Beam selection:

To calculate the preliminary sizing of the steel beams, the beam with the highest bending moment on the roof is calculated as it has the highest imposed load compared to the other floors. Thus, characteristic values obtained for this floor can be assigned to all other floors because the maximum capacity has already been assessed.

First, using the self-weight of the metal decking and the imposed load, the Ultimate Limit State is calculated as,

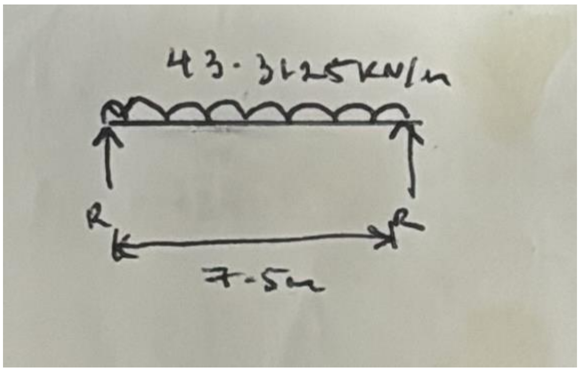
ULS = $1.35G_k + 1.5Q_k$
= 1.35(4.5) + 1.5(7.5)
= 17.325 kN/m^2

Structural model



Using the structural model of the roof floor, we now assess the bending moments of the internal secondary beam and primary beam.

Internal Secondary beam
Length = 7.5m
Span = 2.5m



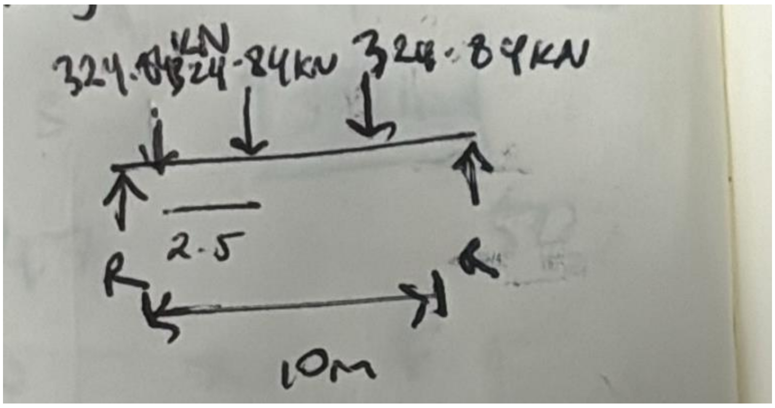
UDL = 2.5 x (17.325)
= 43.3125 kN/m

Reactions = (43.3125 x 7.5)/2
= 162.42 kN

Max BM = $\frac{wl^2}{8}$ = (43.3125 x 7.5^2)/8 = 304.541kNm

Internal Primary beam (supports six internal secondary beams)

Length = 10m



Reactions = 162.42 x 2
= 324.84 kN

Max BM = 2.5/10 x 324.84
= 812.1kNm

From these calculations, the most loaded case comes out to be the internal primary beam with a maximum bending moment equal to 812.1kNm.

Steel section sizing

The steel section size will be chosen based on the internal primary beam by comparing its maximum bending moment to the closest buckling resistance design value $M_{c,y,Rd}$ and axial compression design value $N_{b,y,Rd}$. The dimensions are for Class 1 steel of grade S355. The design buckling resistance value is found using Steel Blue Book for Universal Beam buckling resistance values (for failure span 5.0m).

Section designation Cross-section resistance (kNm) Classification	C_1[1]	Buckling Resistance Moment $M_{b,Rd}$ (kNm) for length between lateral restraints (m)												
		1.0	1.5	2.0	2.5	3.0	3.5	4.0	5.0	6.0	7.0	8.0	9.0	10.0
610 x 178 x 100 +	1.00	963	917	803	701	612	535	470	373	306	258	223	196	176
	1.13	963	959	853	754	665	586	517	411	334	283	245	216	194
$M_{c,y,Rd}$ = 963	1.35	963	963	922	830	744	664	593	475	387	322	280	249	224
$M_{c,z,Rd}$ = 102	1.50	963	963	960	874	791	712	640	518	424	352	303	269	243
	1.77	963	963	963	940	863	789	718	592	491	410	347	304	275
	2.00	963	963	963	963	914	844	776	651	545	460	391	336	301
Class = 1	2.50	963	963	963	963	963	941	880	761	654	562	486	422	369

Figure 4: Buckling-resistant moments of UB Beams

For steel size 610 x 178 x 100, the maximum BM, 812.1kNm, is less than the design buckling resistance moment, 963KNm, meaning it passes the check at a safe value.

Now, to check for the compression resistance at this section size, the axial compression table is used,

Universal beams (UB) Axial compression with S355

Print design data | Export to excel

Section designation		Axis	Compression resistance $N_{b,y,Rd}$, $N_{b,z,Rd}$, $N_{b,T,Rd}$ (kN) for buckling lengths (m)												
			1.0	1.5	2.0	2.5	3.0	3.5	4.0	5.0	6.0	7.0	8.0	9.0	10.0
x 101	*	$N_{b,y,Rd}$	3990	3990	3990	3990	3990	3990	3990	3950	3900	3850	3800	3750	3690
		$N_{b,z,Rd}$	3910	3720	3510	3250	2960	2640	2310	1790	1340	1020	805	649	534
		$N_{b,T,Rd}$	3970	3820	3660	3500	3310	3120	2920	2520	2180	1910	1710	1550	1430
610 x 178 x 100	* +	$N_{b,y,Rd}$	3910	3910	3910	3910	3910	3910	3900	3860	3810	3760	3710	3660	3600
		$N_{b,z,Rd}$	3710	3440	3120	2730	2430	1980	1620	1120	812	613	478	383	314
		$N_{b,T,Rd}$	3810	3620	3410	3190	2960	2730	2520	2150	1890	1700	1570	1470	1400
x 92	* +	$N_{b,y,Rd}$	3590	3590	3590	3590	3590	3590	3580	3540	3500	3450	3400	3350	3300
		$N_{b,z,Rd}$	3400	3140	2820	2440	2160	1760	1430	980	707	533	415	332	272
		$N_{b,T,Rd}$	3490	3310	3120	2900	2680	2450	2230	1880	1620	1440	1320	1230	1160
x 82	* +	$N_{b,y,Rd}$	3080	3080	3080	3080	3080	3080	3080	3040	3010	2970	2930	2880	2840
		$N_{b,z,Rd}$	2910	2690	2410	2080	1730	1490	1210	828	596	449	349	280	229

Figure 5: Axial compression resistances of UB Beams

For the chosen steel section size, the shear force, 324.84kN, is less than the design axial force, which is 3860kN, well within the area of safety. Therefore, for both design values, this section size passes the checks. Thus, all steel beams, including primary beams and secondary beams, will use 610 x 178 x 100 sizes.

Steel Column Sizing

It was decided that one column would be used for all floors. To do this, the column subjected to the highest loads was selected. There are reaction forces from the floor slabs, beams, and imposed loads on the column. Doing so is a cost-effective and time-effective method that does not compromise on safety and structural integrity. The self-weight for the flooring option was extracted from the Bison Manufacturing booklet, and the self-weight of the steel beam was assumed to be 5.0 kN/m.

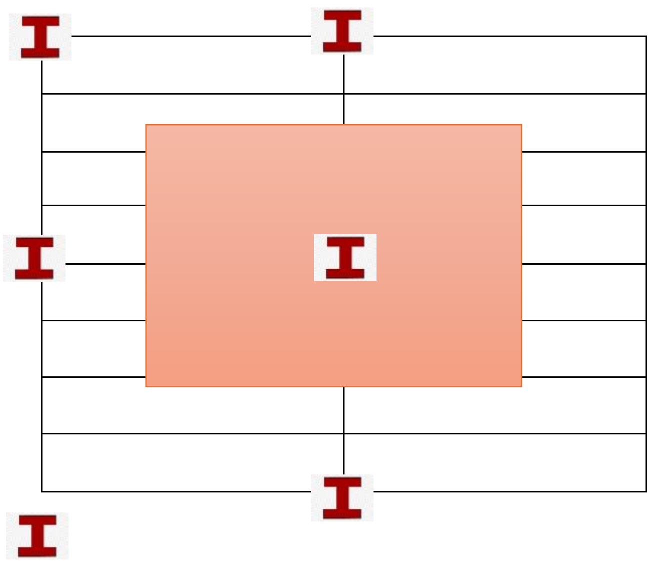


Figure 6: Reaction Forces Acting on the Column

Hollow Composite Floors

Load/Span Table

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			Effective span in metres								
200	150	3.6	8.25	8.25	8.1	7.9	7.7	7.4	7.1	5.8	4.9
250	200	4.2	10.4	9.9	9.7	9.4	9.2	8.8	8.4	7	6
300	250	4.5	11.7	11.2	10.9	10.6	10.4	9.9	9.5	7.8	6.8
375	300	5.8	14.5	14	13.7	13.5	13.2	12.7	12.3	10.7	8.8

Figure 7: Floor slab properties from Bison Manufacturing

Load Analysis Table

Floor Level	Structural Unit / Forced load	Load Magnitude (kN/m^2)	X-Direction	Y-Direction	Total Load Unfactored (kN)	Total Load [1.35x factor] (kN)
Ground	Hollow Composite Floor (250 mm)	4.5	10	10	450	607.5
Ground	Imposed Load	5.0	10	10	500	675
Ground	Services	0.25	10	10	25	33.75
Ground	Beam	0.5 kN/m	10	x	5	6.75
Total						1323 kN
Floor 1	Hollow Composite Floor (250 mm)	4.5	10	10	450	607.5
Floor 1	Imposed Load	4.0	10	10	400	540
Floor 1	Services	0.25	10	10	25	33.75
Floor 1	Beam	0.5 kN/m	10	x	5	6.75
Total						1188 kN
The same structural units and load(s) will be present in Floor 2 and Floor 3, so total factored load should be multiplied by 3.						
Total (Floor 1-3)						3564 kN
Roof	Hollow Composite Floor (250 mm)	4.5	10	10	450	607.5
Roof	Imposed Load	7.5	10	10	750	1012.5
Roof	Beam	0.5 kN/m	10	x	5	6.75
Total						1626.75 kN

For the sizing of the steel columns, only the axial loads for floors 1 to 4 and the roof that will be considered because the ground floor's axial load is absorbed by the building foundations and hence dismissed. Hence,

$$Total\ Axial\ Load = 5190.75\ kN$$

By Re-arranging the compression resistance formula, we can solve for the cross-sectional area A and then use it to find the appropriate column size. The same column will be constant and used for all floors.

$$A = \frac{N_{c,Rd} \times \gamma_{M0}}{F_y}$$

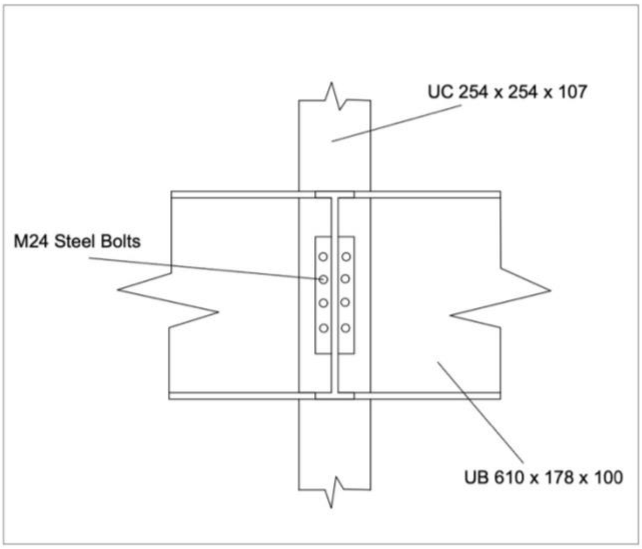
$$A = \frac{5190.75 \times 10 \times 1.05}{355} = 153.529\ cm^2$$

254 x 254	**x 167**	2080	744	2420	1140	0.851	8.48	1.63	626	213
	x 132	1630	576	1870	878	0.850	10.3	1.19	319	168
	x 107	1310	458	1480	697	0.848	12.4	0.898	172	136
	x 89	1100	379	1220	575	0.850	14.5	0.717	102	113
	x 73	898	307	992	465	0.849	17.2	0.562	57.6	93.1

Figure 8: Column Size and section properties

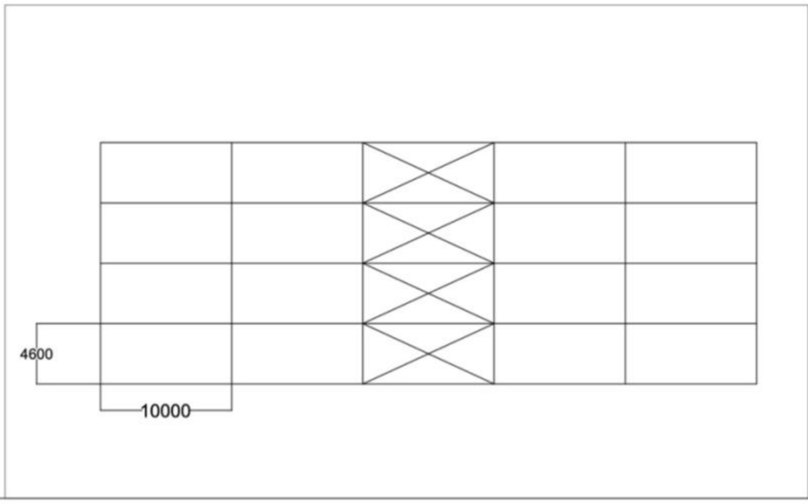
UC254x254x132 Is an appropriate column size for this project.

4: AutoCAD Drawings



Section View, Scale:1:50

Side Plan View, Scale:1:50



Notes:

The section view describes the connection between the Universal Beam & Universal Column using M24 Bolts. Furthermore, the Side view shows the steel frame (our final pick) and its number of storey's as well as the vertical cross-bracing in the middle. Its length is 4.6m each storey and a length of 10m in between the 5 columns.

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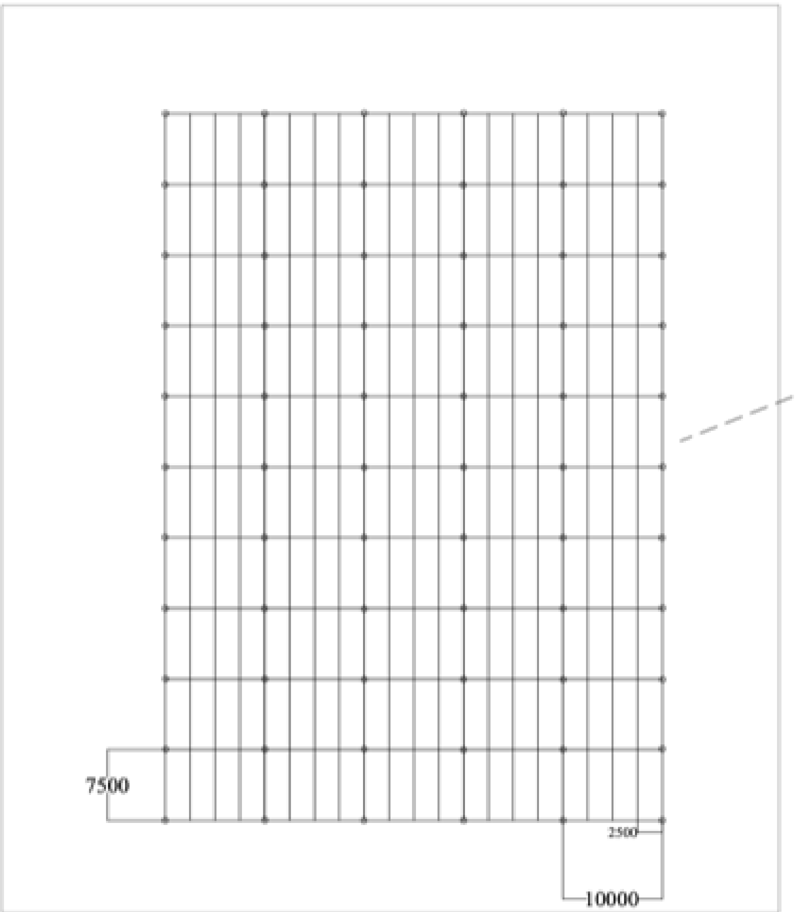

Floor Plan View, Scale: 1:50

Cross-Bracing in the centres of the Frame

Notes:

This Floor plan view demonstrates the positioning of components such slabs, beams, and columns This is also showing how the Universal primary and secondary beams are arranged, how the composite metal decking slabs are laid out, and how far apart the columns are spaced (7.5 meters). This view also shows the placement of the cross-bracing which is in the centre as shown by the red lines.

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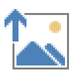
5.1 : Method of construction

Contractor's responsibility :

All work during construction will be managed by the contractor, and the contractor will be allowed to distribute different tasks to different subcontractors. The supplier will manage the transportation of materials, and the trade contractor will manage the budget of the project, including the procurement of materials and the wages of the workers, etc. In addition, to ensure the professionalism and feasibility of the project and the sustainable use of the building, the contractor should also allow professional technical institutions to intervene to ensure the theoretical correctness。

Stage	Tasks	Materials and Equipment	Stakeholders responsible	Expected duration	Health and safety References
1.01- Before construction	The health and safety executive must be notified about the start of work and construction. To ensure the construction process adheres to safety protocols.	Barriers to make people aware 	Project manager Health and safety executive	1 day	CDM 2015 Reg. 5, 15 Health and Safety at work
1.02	Engineering managers need to carry out very detailed task allocation and time management to ensure the normal operation of the project and complete in set time	Timetable	Project Manager	1 day	N/A
1.03	The supplier of the materials should choose the supplier closest to the building in order to avoid transportation delays caused by traffic jams. In addition, the transportation time of materials should be strictly monitored to ensure that sufficient time is allowed in case of accidents	Flatbed trailer 	Project Manager Driver	4 days	CDM 2015 Reg. 8 Road Traffic Regulation Act; Vehicle and Operator Services Agency Regulations
1.04	Before the foundation construction the land should be surveyed, including bearing strength and land levelling.	Land survey machine 	Geotechnics Construction Workers	2 days	CDM 2015 Reg. 22 Control of Substances Hazardous to Health Regulations 2002
1.05	The welding positions where to fix the metal decking should be clearly marked in the early stage。 To avoid any curing on the side of metal plate， mild steel cable should be applied for liftin。	welding equipment 	Geotechnics construction workers	1 day	CDM 2015 Reg.6, 27 Health and Safety at Work Act

	Meanwhile, effective measures should be taken to avoid any leakage when pouring the concrete				
2.01- During Construction	Ground floor columns secured, and concrete is poured.	Plain concrete cement Crane 	Structural engineers Construction Workers	2 weeks	CDM 2015 Reg.15, 16
2.02	Primary beams connected to ground floor columns using weld plates.	7.5 meters long for primary beams 450-600 mm deep for primary beams Cranes 	Contractors Construction Workers Welders	2 weeks	CDM 2015 Reg. 18
2.03	Secondary beams connected to the primary beams.	4.17 meters long for the secondary beams Cranes 	Contractors Construction Workers	2 weeks	CDM 2015 Reg. 18
2.05	Composite metal decking is connected to beams It must has reasonable span between beams to ensure the slab thickness and deflection.	Spans of 4-5 meters for metal decking Slab thickness of 130-150 mm for metal decking	Contractors Construction Workers	1 week	CDM 2015 Reg. 18

6: Refurbishment and Addition of a Floor to the Existing 5-storey RC Building

6.1: Introduction

To refurbish and extend a five-story reinforced concrete office building from the 1980s, structural issues must be resolved, current codes must be followed, and extra loads must be kept to a minimum while preserving the building's structural integrity. With an emphasis on structural reinforcement, material selection, and regulatory issues, this section addresses how the current concrete structure can be renovated and the strategy for building a new sixth level.

6.2 Structural Assessment and Strengthening

A thorough structural examination must be carried out before the start of any renovation or additional projects. This will entail assessing the state of the slabs, beams, and columns made of reinforced concrete. The incorporated steel reinforcement in concrete structures from the 1980s may corrode because of carbonation, chloride intrusion, or cracking. Before granting any extension, these problems must be resolved.

Concrete Deterioration: Non-destructive testing (NDT) techniques, like ultrasonic pulse velocity tests or rebound hammer tests, can be used to determine the strength of the concrete without causing any damage, therefore determining the level of concrete degradation. Concrete patching for locations where the concrete has deteriorated or spalled, or epoxy injections for crack filling, can be used to fix any damage found. It is also possible to use cathodic protection to prevent the steel reinforcement from corroding.

Reinforcement and Concrete Repair: Concrete jacketing might be used if the building's structural integrity has been jeopardised because of corroded reinforcement or weakening concrete. To increase the load-bearing capability of existing beams and columns, a fresh coating of reinforced concrete is applied around them. Furthermore, any exposed reinforcement or spalled concrete needs to be cleaned, treated with anti-corrosion chemicals and replaced with high-strength concrete.

Foundation Underpinning: When a new floor is added, one of the main questions is whether the current foundation can support the extra weight. If the foundation has to be strengthened, it will be determined via a load evaluation. The current foundation can be strengthened if necessary by using underpinning techniques like mini-piling or micropyles. Micropyles are useful because they can support heavy loads and can be added with little interference with the current structure.

6.3: Including a Floor

Selecting building materials and techniques for the new sixth storey that won't put too much strain on the existing structure is crucial because the old structure was constructed more than 40 years ago. Because of its strength and relatively lightweight in comparison to concrete, steel is the perfect material for this extension.

Lightweight Steel Frame: A lightweight steel frame will support the new sixth storey. This enables quick installation while reducing the extra strain on the concrete framework. Prefabricated steel beams and columns can be swiftly constructed on-site after being made off-site, cutting down on construction time and causing the least amount of disturbance to building occupants. The new floor will provide the required support without putting undue strain on the lower floors because of the steel's strong strength-to-weight ratio.

Composite Metal Decking: Composite metal decking is suggested for the flooring system. Using this technique, concrete is poured on top of metal decking that has been placed over steel beams. Compared to conventional in-situ concrete floors, this results in a lightweight, robust floor structure that is simpler to manufacture. Additionally, the decking makes it simple to integrate HVAC ducts, plumbing, and electrical cabling between the steel beams.

Load Transfer: The existing reinforced concrete structure and the steel frame need to be carefully integrated. This entails joining the new floor's steel columns to the concrete columns underneath with bolted or welded joints. This uniformly distributes the weight from the new floor to the foundation. The load transfer points need to be carefully considered to avoid localised stress concentrations that can cause failure or cracking.

6.4: Wind loads and lateral stability

The building becomes higher as a new floor is added, increasing its vulnerability to wind loads. Additional lateral stability measures must be included to guarantee the building's stability, especially against lateral stresses like wind or seismic activity.

Shear Walls and Cross Bracing: Steel cross bracing or shear walls can be erected to withstand wind stresses. Although concrete buildings are more likely to use shear walls, steel cross-bracing is lighter and offers superior lateral stability without significantly increasing the structure's weight. To ensure that lateral pressures are dispersed throughout the building, the cross-bracing will be built to join the steel frame of the new floor with the existing structure.

Foundation Strengthening for Stability: To avoid uplift or excessive lateral movement, further foundation strengthening may be necessary in situations where strong wind loads are anticipated. To make sure the building can withstand greater lateral stresses from the taller structure, deepened footings or pile foundations may be utilised.

6.5 Adherence to Building Regulations

It is crucial to make sure that the renovated building conforms with current UK construction rules, including updated fire safety and energy efficiency standards because the original structure was constructed in the 1980s.

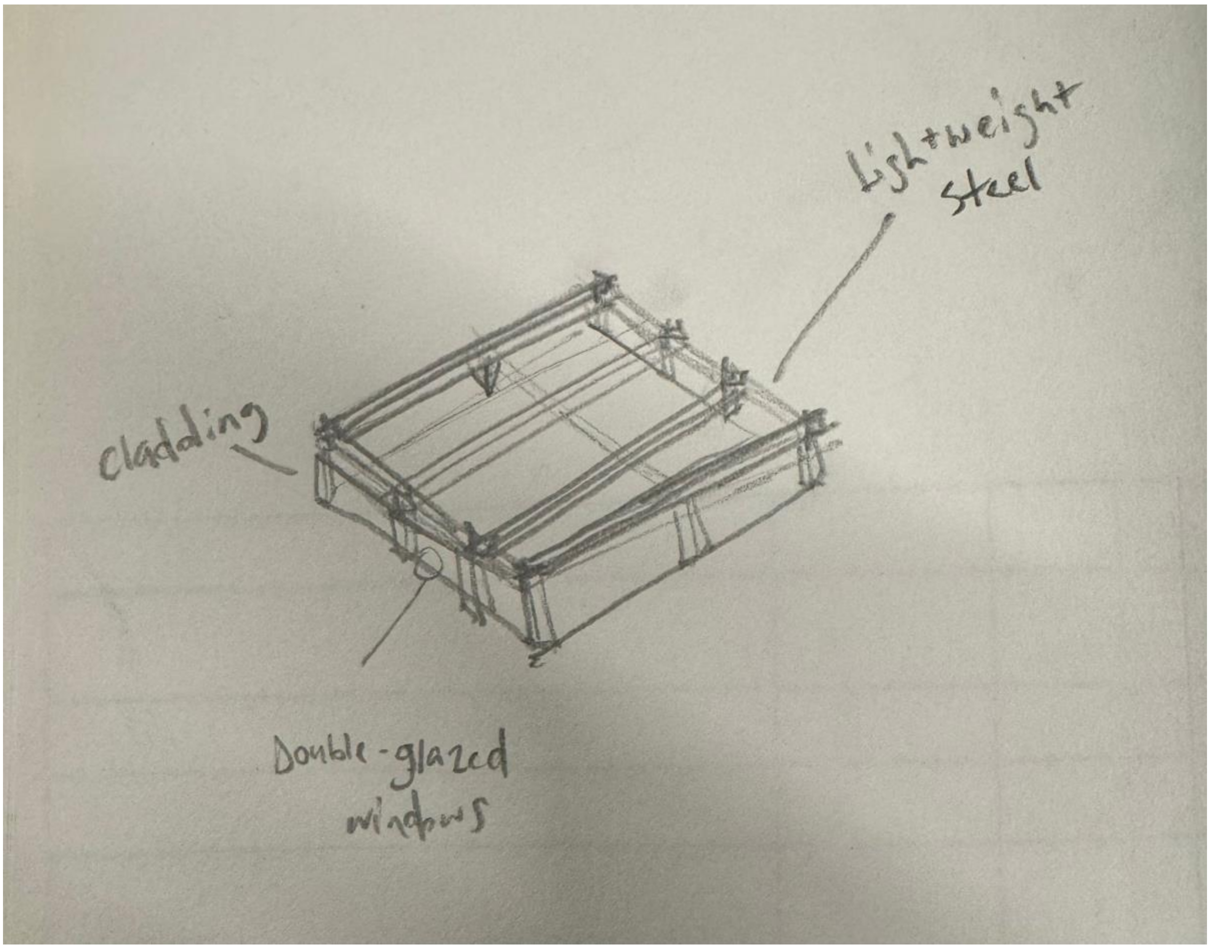
Fire Safety: The building must adhere to the Fire Safety Act 2021, which may call for changes to fire escape routes and evacuation protocols, as well as fire-resistant coating on the new steel frame. To guarantee that the steel frame retains its structural integrity in the case of a fire, fireproof coatings ought to be put on it.

Energy Efficiency: Improving the building's energy efficiency is another benefit of renovation. Modernising the HVAC system, installing double-glazed windows, and improving insulation will all help the building use less energy and satisfy energy performance standards.

6.6: Conclusion

Strengthening the current structure to handle any degradation and making sure the new sixth level is lightweight and well-integrated are two aspects of the renovation and expansion of the five-story office building. The added load on the existing foundation is reduced by employing composite metal decking and a lightweight steel frame. Cross-bracing or shear walls are used to provide lateral stability, and the renovated structure's safety and energy efficiency are guaranteed by adherence to contemporary building codes.

Hand drawing illustrating the top floor and its components as was described above:



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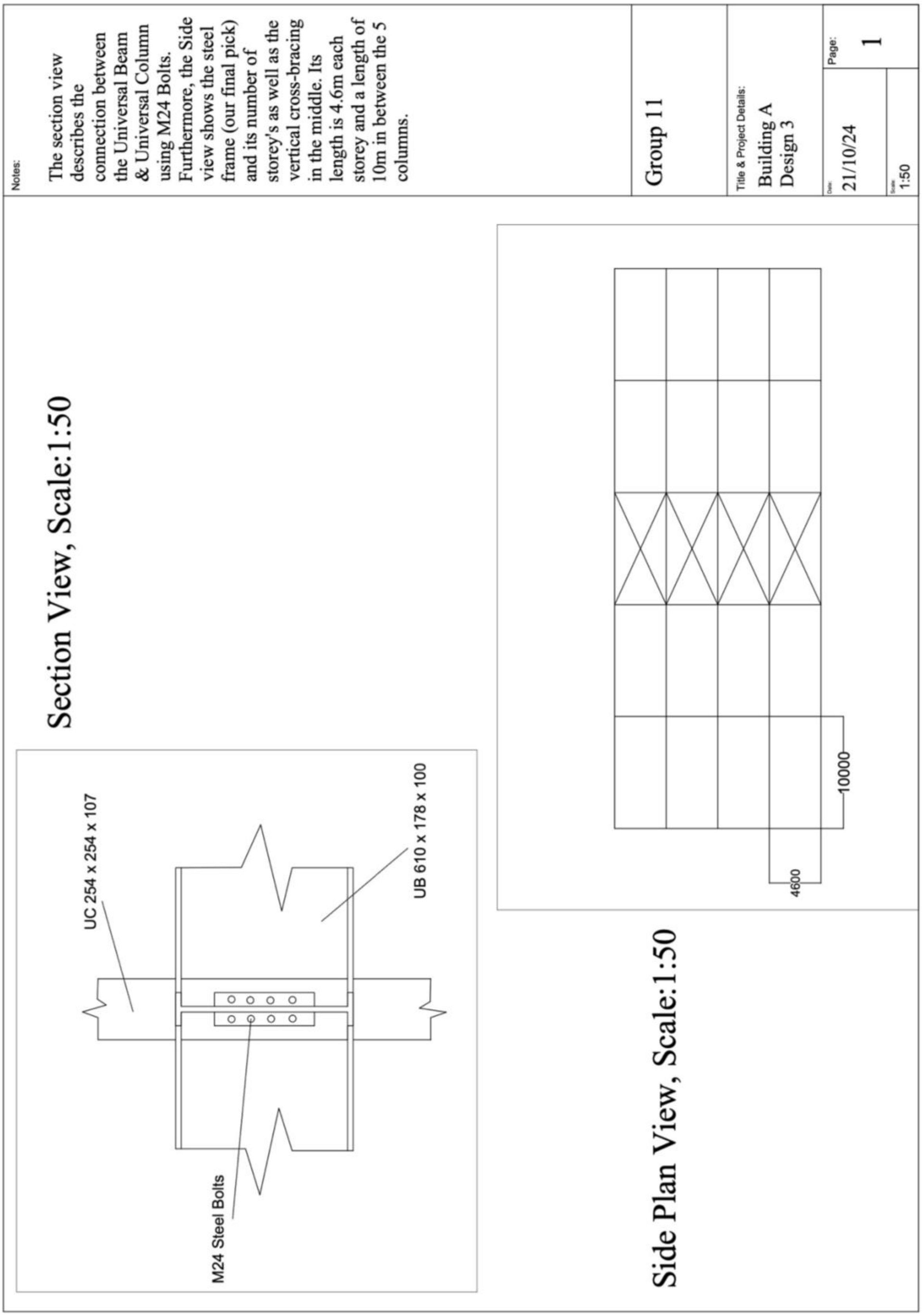
Appendix


Section View, Scale:1:50
UC 254 x 254 x 107
UB 610 x 178 x 100
M24 Steel Bolts
Side Plan View, Scale:1:50
10000
4600
Notes:
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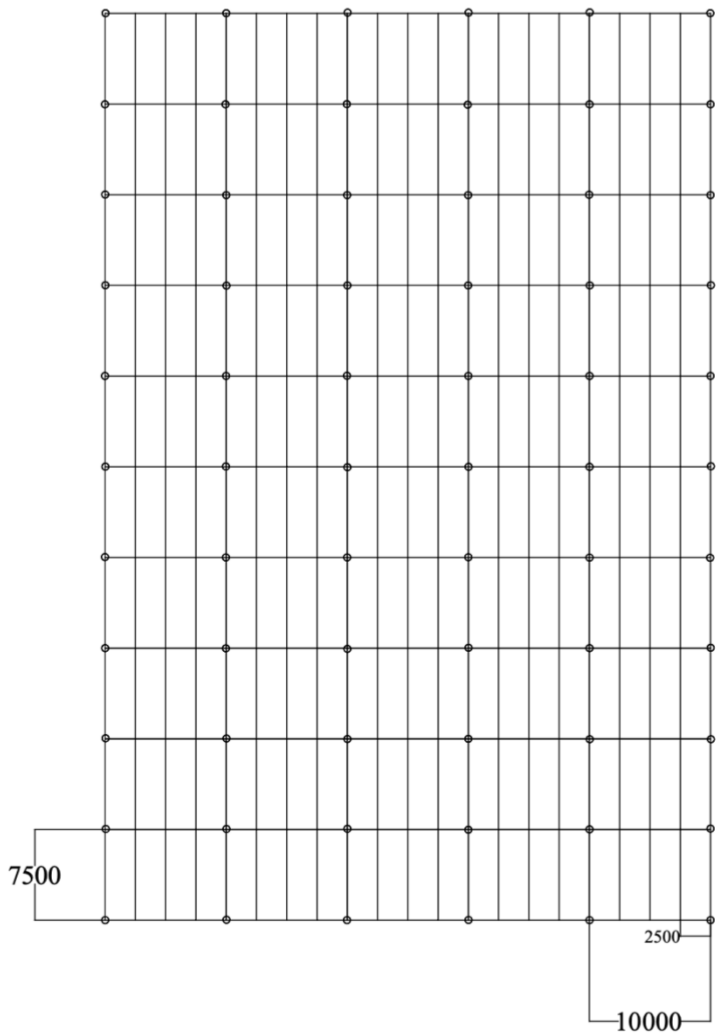
Floor Plan View, Scale: 1:50



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